

Bohloko Family Farm Poultry Processing System - Use Case Document

1. Introduction

1.1 Purpose

This document outlines the functional requirements of the Bohloko Family Farm Poultry Processing System through use cases. The system is designed to manage the complete poultry production, processing, inventory, sales, and distribution operations for Bohloko Family Farm.

1.2 System Overview

The Bohloko Family Farm Poultry Processing System is a comprehensive software solution that supports:

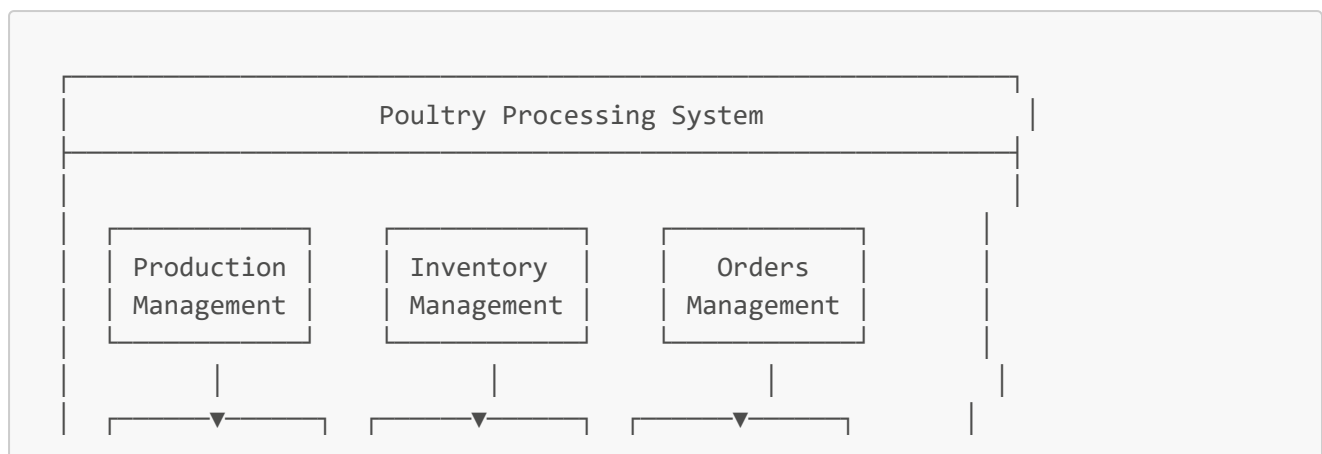
- **Production Management:** Tracking broiler cycles, feed consumption, medication, and mortality
- **Inventory Management:** Managing processed chicken products, storage, and stock levels
- **Order Processing:** Handling customer orders from various channels
- **User Management:** Role-based access control for farm staff and external customers
- **Analytics & Reporting:** Business intelligence for decision making
- **Compliance Tracking:** Food safety and regulatory compliance documentation

1.3 Actors

1. **Farm Manager** - Overall system administrator with full access
2. **Poultry Attendant** - Manages live bird production and daily operations
3. **Processing Staff** - Handles slaughtering, processing, and inventory management
4. **Sales Assistant** - Manages customer orders and sales
5. **Customer** - External users who place orders (Consumer, Restaurant, Retailer, Distributor, Farm Gate, Institution)
6. **System Administrator** - Technical system maintenance

2. Use Case Diagrams

2.1 High-Level Use Case Diagram



Plan Cycle	Track Stock	Place Order		
Record Feed	Update Levels	Process Order		
Log Mortality	Manage Batch	Track Status		

3. Detailed Use Cases

3.1 Authentication & Authorization

UC-001: User Registration

- **Actor:** Customer, Staff Member
- **Description:** New users can register for an account with the system
- **Preconditions:** User does not have an existing account
- **Basic Flow:**
 1. User navigates to registration page
 2. User selects user type (Consumer, Restaurant, Retailer, etc.)
 3. User provides required information (email, password, business details)
 4. System validates information and creates account
 5. System sends confirmation email
 6. Account status is set to "pending" until approved by Farm Manager
- **Postconditions:** User account created with pending status
- **Extensions:**
 - 3a. Email already exists: System displays error message
 - 3b. Invalid business registration: System prompts for correction

UC-002: User Login

- **Actor:** All authenticated users
- **Description:** Users log into the system with credentials
- **Preconditions:** User has an approved account
- **Basic Flow:**
 1. User enters email and password
 2. System validates credentials
 3. System checks account status and lock status
 4. System generates authentication token
 5. User is redirected to appropriate dashboard based on role
- **Postconditions:** User session established
- **Extensions:**
 - 2a. Invalid credentials: System increments login attempts, locks after 5 failures
 - 2b. Account not approved: System displays pending approval message

3.2 Production Management

UC-003: Plan Production Cycle

- **Actor:** Farm Manager, Poultry Attendant
- **Description:** Create a new broiler production cycle plan
- **Preconditions:** User has appropriate role permissions
- **Basic Flow:**
 1. User navigates to Production Planning
 2. User creates new cycle with details (type, expected birds, duration)
 3. User sets budget for feed, medication, labor, utilities
 4. System validates plan and saves as "planned"
 5. Farm Manager approves the plan
 6. System creates production cycle with "in_progress" status
- **Postconditions:** New production cycle created and active
- **Extensions:**
 - 5a. Plan rejected: Farm Manager provides rejection reason

UC-004: Record Daily Production Log

- **Actor:** Poultry Attendant
- **Description:** Record daily observations and activities for active production cycle
- **Preconditions:** Active production cycle exists
- **Basic Flow:**
 1. User selects active production cycle
 2. User creates daily log entry
 3. User records birds count, feed consumed, water consumed, mortality
 4. User records environmental conditions (temperature, humidity)
 5. User logs daily activities and any issues
 6. System saves log and updates cycle statistics
- **Postconditions:** Daily production data recorded
- **Extensions:**
 - 3a. High mortality detected: System triggers alert to Farm Manager

UC-005: Record Medication & Vaccination

- **Actor:** Poultry Attendant
- **Description:** Log medication and vaccination administration
- **Preconditions:** Active production cycle exists
- **Basic Flow:**
 1. User selects active production cycle
 2. User adds medication/vaccination record
 3. User specifies medication name, dosage, date, administered by
 4. System validates and saves record
 5. System updates medication schedule
- **Postconditions:** Medication history maintained for compliance

3.3 Inventory Management

UC-006: Process Harvested Birds

- **Actor:** Processing Staff
- **Description:** Convert live birds to processed inventory
- **Preconditions:** Production cycle ready for harvest
- **Basic Flow:**
 1. User selects production cycle for harvest
 2. User records number of birds harvested
 3. System creates inventory batches for different cuts (whole, portions)
 4. User assigns batch numbers, weights, and storage locations
 5. System updates inventory levels and marks cycle as completed
- **Postconditions:** Inventory increased, production cycle completed

UC-007: Manage Inventory Levels

- **Actor:** Processing Staff, Farm Manager
- **Description:** Monitor and update inventory stock levels
- **Preconditions:** Inventory items exist in system
- **Basic Flow:**
 1. User views current inventory dashboard
 2. System displays stock levels, expiry dates, storage locations
 3. User can adjust quantities (add/remove/damage)
 4. User can transfer between storage locations
 5. System updates real-time inventory counts
- **Postconditions:** Accurate inventory tracking maintained

UC-008: Generate Inventory Reports

- **Actor:** Farm Manager, Processing Staff
- **Description:** Create inventory status and valuation reports
- **Preconditions:** Inventory data exists
- **Basic Flow:**
 1. User selects report type (stock levels, valuation, expiry)
 2. User sets date range and filters
 3. System generates report with totals and trends
 4. User can export to PDF/Excel
- **Postconditions:** Inventory insights available for decision making

3.4 Order Management

UC-009: Place Order

- **Actor:** Customer (Consumer, Restaurant, Retailer, etc.)
- **Description:** Customer places order for poultry products
- **Preconditions:** Customer account approved, inventory available
- **Basic Flow:**
 1. Customer browses product catalog
 2. Customer adds items to cart with quantities
 3. Customer selects delivery/pickup option

- 4. System validates inventory availability
- 5. Customer confirms order and payment
- 6. System creates order with "pending" status
- **Postconditions:** Order created awaiting processing
- **Extensions:**
 - 4a. Insufficient inventory: System suggests alternatives or backorder

UC-010: Process Order

- **Actor:** Sales Assistant
- **Description:** Process customer orders through fulfillment
- **Preconditions:** Order exists in "pending" status
- **Basic Flow:**
 1. User views pending orders queue
 2. User selects order for processing
 3. System reserves inventory for order
 4. User confirms picking/packing
 5. User updates order status to "processing" then "shipped"
 6. System sends notifications to customer
- **Postconditions:** Order fulfilled, inventory reduced

UC-011: Track Order Status

- **Actor:** Customer, Sales Assistant
- **Description:** Monitor order progress through fulfillment stages
- **Preconditions:** Order exists in system
- **Basic Flow:**
 1. User navigates to order tracking
 2. System displays current status and timeline
 3. For customers: View estimated delivery
 4. For staff: Update status and add notes
 5. System logs all status changes
- **Postconditions:** Order transparency maintained

UC-012: Process Payment

- **Actor:** Customer, Sales Assistant
- **Description:** Handle payment for customer orders
- **Preconditions:** Order exists with "pending" payment status
- **Basic Flow:**
 1. Customer selects payment method (credit card, bank transfer, mobile money, cash)
 2. System processes payment through appropriate gateway
 3. Payment gateway returns transaction status
 4. System updates order payment status to "paid" if successful
 5. System generates payment receipt
 6. Order status moves to "confirmed"

- **Postconditions:** Order payment completed, order confirmed
- **Extensions:**
 - 3a. Payment fails: System retries or prompts for alternative method
 - 3b. Partial payment: System tracks partial payment status

UC-013: Cancel Order

- **Actor:** Customer, Sales Assistant, Farm Manager
- **Description:** Cancel an existing order with appropriate refunds
- **Preconditions:** Order exists and is cancellable (pending or confirmed status)
- **Basic Flow:**
 1. User requests order cancellation with reason
 2. System checks if order can be cancelled (not shipped)
 3. If payment was made, system initiates refund process
 4. System updates order status to "cancelled"
 5. System releases reserved inventory back to stock
 6. Customer receives cancellation confirmation and refund if applicable
- **Postconditions:** Order cancelled, inventory restored, refund processed if needed

3.5 User & Role Management

UC-014: Manage User Accounts

- **Actor:** Farm Manager
- **Description:** Approve, suspend, or manage user accounts
- **Preconditions:** User has Farm Manager role
- **Basic Flow:**
 1. User views pending account approvals
 2. User reviews business registration and details
 3. User approves or rejects account
 4. If rejected, provides reason
 5. System updates account status and notifies user
- **Postconditions:** User account status updated

UC-015: Assign User Roles

- **Actor:** Farm Manager
- **Description:** Assign or modify user roles and permissions
- **Preconditions:** User account exists and is approved
- **Basic Flow:**
 1. User searches for existing user
 2. User views current role and permissions
 3. User assigns new role (Poultry Attendant, Processing Staff, etc.)
 4. System updates permissions and notifies user
- **Postconditions:** User access rights updated

3.6 Analytics & Reporting

UC-016: View Production Analytics

- **Actor:** Farm Manager
- **Description:** Analyze production performance metrics
- **Preconditions:** Production data exists
- **Basic Flow:**
 1. User navigates to Analytics dashboard
 2. System displays key metrics (mortality rate, feed conversion, cycle duration)
 3. User can filter by date range, cycle type
 4. System generates charts and trends
 5. User can drill down to individual cycles
- **Postconditions:** Production insights available

UC-017: Generate Financial Reports

- **Actor:** Farm Manager
- **Description:** Create profit/loss, revenue, and cost reports
- **Preconditions:** Financial data exists
- **Basic Flow:**
 1. User selects report type (P&L, revenue by channel, cost analysis)
 2. User sets reporting period
 3. System calculates totals and margins
 4. System generates report with comparisons to previous periods
 5. User can export for accounting purposes
- **Postconditions:** Financial performance documented

3.7 Compliance Management

UC-018: Record Compliance Checks

- **Actor:** Processing Staff, Farm Manager
- **Description:** Document food safety and quality compliance checks
- **Preconditions:** Processing activities occurring
- **Basic Flow:**
 1. User performs quality check on batch
 2. User records check results (pass/fail) in system
 3. User documents any corrective actions
 4. System timestamps and stores compliance record
 5. System generates compliance certificates if needed
- **Postconditions:** Compliance documentation maintained

UC-019: Generate Compliance Reports

- **Actor:** Farm Manager
- **Description:** Create reports for regulatory authorities
- **Preconditions:** Compliance data exists
- **Basic Flow:**

1. User selects compliance report template
 2. System compiles all compliance records for period
 3. System formats report according to regulatory requirements
 4. User reviews and approves report
 5. System generates final document for submission
- **Postconditions:** Regulatory compliance documented

4. Non-Functional Requirements

4.1 Performance Requirements

- System should support 50 concurrent users
- Page load times under 3 seconds
- Order processing within 30 seconds
- Report generation under 60 seconds for 12-month period

4.2 Security Requirements

- Role-based access control
- Password encryption (bcrypt)
- HTTPS for all communications
- Audit logging for all critical operations
- Regular security updates and patches

4.3 Availability Requirements

- 99.5% uptime during business hours (6 AM - 10 PM)
- Automated daily backups
- Disaster recovery plan with 4-hour RTO

4.4 Usability Requirements

- Responsive design for mobile and desktop
- Intuitive navigation for users with varying tech literacy
- Multi-language support (English, Sesotho initially)
- Accessibility compliance (WCAG 2.1 AA)

5. System Constraints

5.1 Technical Constraints

- Must work with existing farm infrastructure (limited internet in rural areas)
- Offline capability for critical operations
- Mobile-first design for field staff
- Integration with existing payment systems (cash, EFT, mobile money)

5.2 Business Constraints

- Must comply with South African food safety regulations

- Must support multiple sales channels (farm gate, online, wholesale)
- Must accommodate seasonal production variations
- Must support growth from 500 to 5000+ birds per cycle

6. Glossary

- **Broiler Cycle:** 5-6 week production period from chick to harvest
- **Feed Conversion Ratio:** Kilograms of feed per kilogram of weight gain
- **Mortality Rate:** Percentage of birds that die during production cycle
- **Batch:** Group of processed chickens with same harvest date and characteristics
- **User Types:** Consumer, Restaurant, Retailer, Distributor, Farm Gate, Institution
- **Production Status:** Planned, In Progress, Completed, Cancelled

7. Appendix

7.1 User Role Matrix

Role	Production	Inventory	Orders	Users	Analytics	Settings
Farm Manager	Full	Full	Full	Full	Full	Full
Poultry Attendant	Create/Read	Read	None	None	Read	None
Processing Staff	Read	Create/Update	Read	None	Read	None
Sales Assistant	None	Read	Create/Update	None	Read	None
Customer	None	Read	Own orders	Own profile	None	Own settings

7.2 Data Flow Diagrams

(To be developed during detailed design phase)

7.3 Screen Mockups

(To be developed during UI/UX design phase)

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